LevelUp | Construction Job Demand

INTRODUCTION: For this (completely optional) LevelUp Milestone, you'll play the role of a Junior Data Analyst on the Data Team at Hover, a company that uses software to map 3-D models of homes. Your manager is interested in the demand of jobs. She has a hunch that severe weather events have an effect on the number of job requests, but she'd like you to help her prove her hunch is true with data.

You'll be using subqueries to analyze data regarding construction jobs and the role weather might play in driving future demand. Companies in this space need to be able to assess what future demand will look like in order to allocate resources and teams accordingly. Sending a small team to a location may mean potential customers will go with a competitor because they can get the job done in a shorter time frame due to your limited resources.

HOW IT WORKS: Follow the prompts in the questions below to investigate your data. Post your answers in the provided boxes: the **yellow boxes** for the queries you write, and **blue boxes** for text-based answers. When you're done, export your document as a pdf file and submit it on the Milestone page – see instructions for creating a PDF at the end of the Milestone. Please don't *ever* remove (paste your query below 👇) or (write your **answer** below 👇). These help your Evaluator!

– Data Set **Description**

The data needed for this Milestone (`hover.*`) comes from two different sources.

The first data source (`jobs`) consists of historical data from a software solution provider for construction companies:

- **`job_identifier`** - The unique ID for a job
- **`organization_id`** - ID for the account that purchased the job
- **`job_location_city`** - The city where the job was located
- **`job_location_region_code`** - The state where the job was located

- **`job_first_upload_complete_datetime`** - The date on which the customer uploaded photos for the job.
- **`job_deliverable`** - the type of job that the customer requested. Either complete (full building) or roof.

The second dataset (`weather`) comes from the NOAA Storm Prediction Center and catalogs adverse weather events across the United States:

- **`comments`** - A description of the weather event
- **`county`** - The county of the weather event
- **`state`** - The state of the weather event
- **`location`** - The address of the weather event
- **`longitude`** - The longitude of the weather event
- **`latitude`** - The latitude of the weather event
- **`datetime`** - The datetime of the weather event
- **`composite_key`** - Unique key for the weather event, combining the timestamp, longitude, and latitude of the event

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– **Task 1:** Explore the jobs data.

Your manager would like to find out more information regarding when customers make requests.

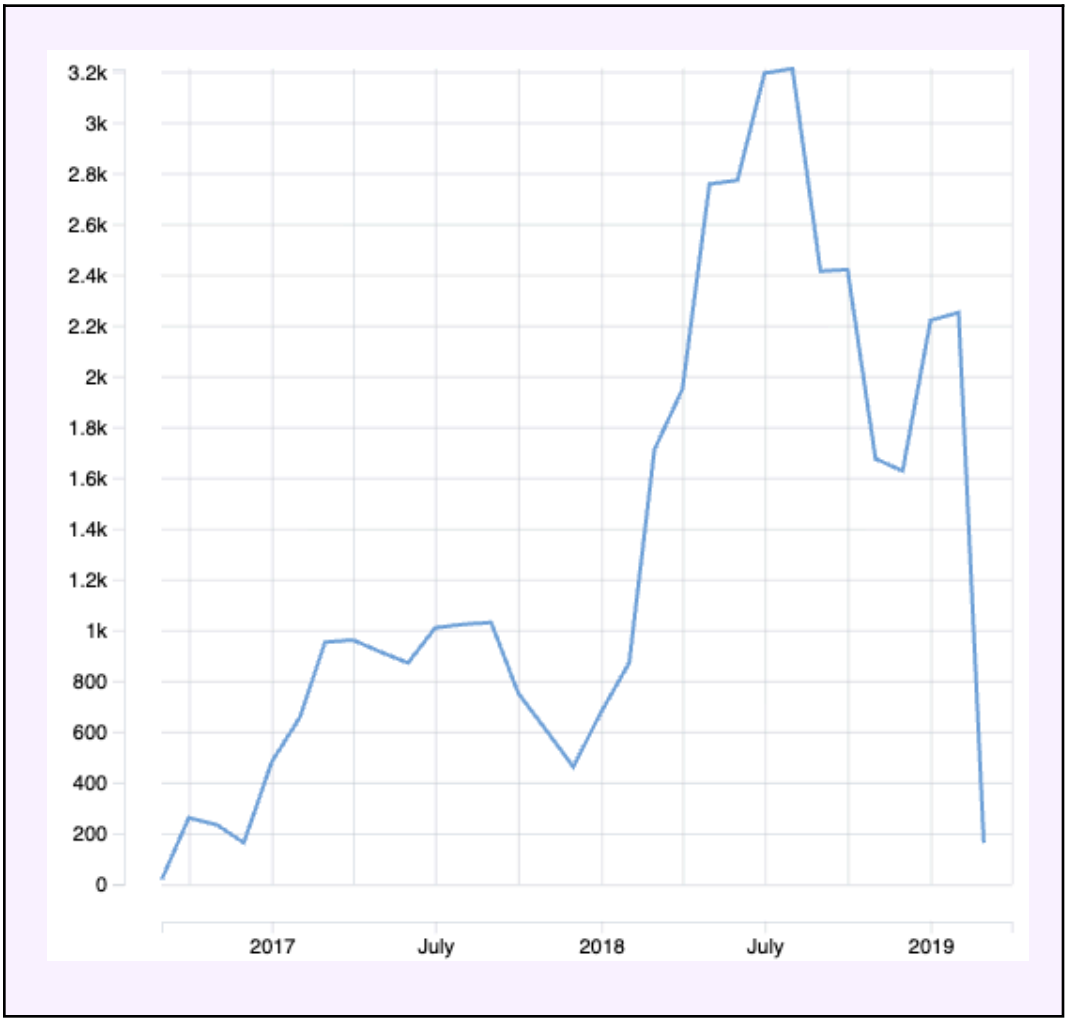
A. Write a query that returns the total number of jobs at the monthly level for each year (i.e. Sept 2016, Oct 2016, etc.). You'll need to use the `date_trunc` function in SQL with the **`job_first_upload_complete_datetime`** variable. When done correctly, your output will have 31 rows.

(paste your query below 👇)

```
SELECT
  date_trunc('month', job_first_upload_complete_datetime) AS
monthly,
  COUNT(*) AS num_jobs
FROM hover.jobs
```

```
GROUP BY
  monthly
ORDER BY
  monthly
```

B. Use the SQL app's built-in visualizer to graph the task in part A.



Is there any seasonality to the job requests? Seasonality can be described as a pattern that repeats itself over cycles of time, such as yearly. Is there a season that has more requests than others?

(write your **answer** below 👇)

Yes, it seems the number of job requests sent out is seasonal. We can see the low points during the winter season (Jan - Feb) and the peak points during the summer season (May to September). This suggests a seasonal trend as it repeats every year but there is also a yearly factor where yearly the job requests increase.

C. Level Up[2] - One of the reasons why we didn't look at data aggregated by month (without the year) is because the first month of the data is September 2016 and the last month of data is March 2019, so every month wouldn't have appeared an equal number of times in a `GROUP BY`. Can you write a query that counts how many times each month has appeared in the data? Hint: You'll need both the date_part and the date_trunc function here. You should `GROUP BY` month, using the date_part function. If done correctly, each month should have a count of 2 or 3.

(paste your query below 👇)

```
SELECT
  date_part('month', job_first_upload_complete_datetime) AS
months,
  COUNT(DISTINCT(date_trunc('month',
job_first_upload_complete_datetime))) AS num_times
FROM hover.jobs
GROUP BY
  months
ORDER BY
  months
```

— **Task 2:** Explore the weather data.

In Task 1, you discovered that there are seasonal patterns to when customers submit job requests. Now you'll investigate the weather data and see how that can help you further with your analysis.

A. The entire weather dataset consists of “adverse weather events”, e.g., tornados, fallen trees, sustained high gusts of wind, etc. Write a query that counts the number of adverse weather events for each month and year of the data (i.e. Sept 2016, Oct 2016, etc.).

(paste your query below 👇)

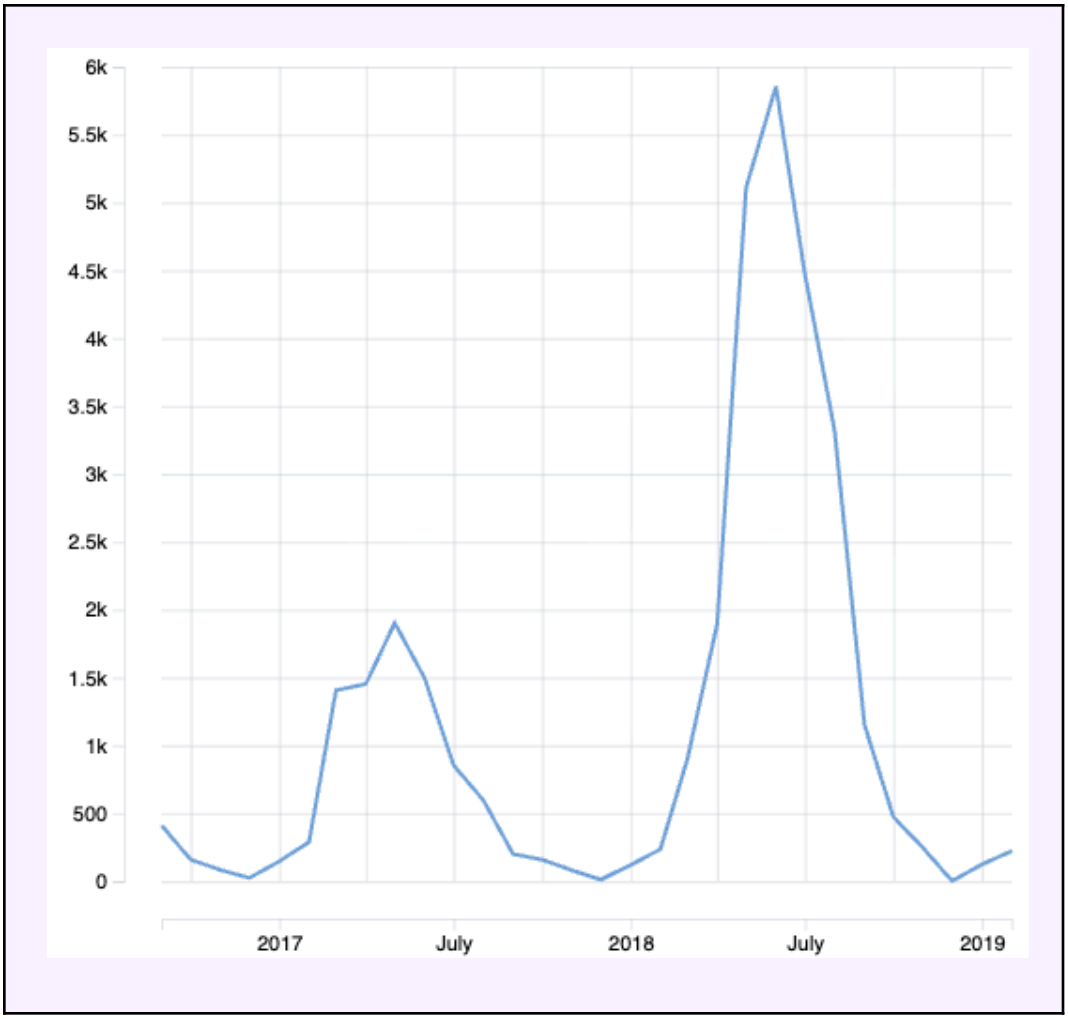
```
SELECT
  date_trunc('month', datetime) AS monthly,
  COUNT(*) as num_weather_events
FROM hover.weather
GROUP BY
  monthly
ORDER BY
  monthly
```

B. Modify your query in 2A to filter out any information prior to September 2016 (the start of the job request data).

(paste your query below 👇)

```
SELECT
  date_trunc('month', datetime) AS monthly,
  COUNT(*) as num_weather_events
FROM hover.weather
WHERE
  date_trunc('month', datetime) >= '2016-09-01'
GROUP BY
  monthly
ORDER BY
  monthly
```

Visualize the data using the built-in visualizer.



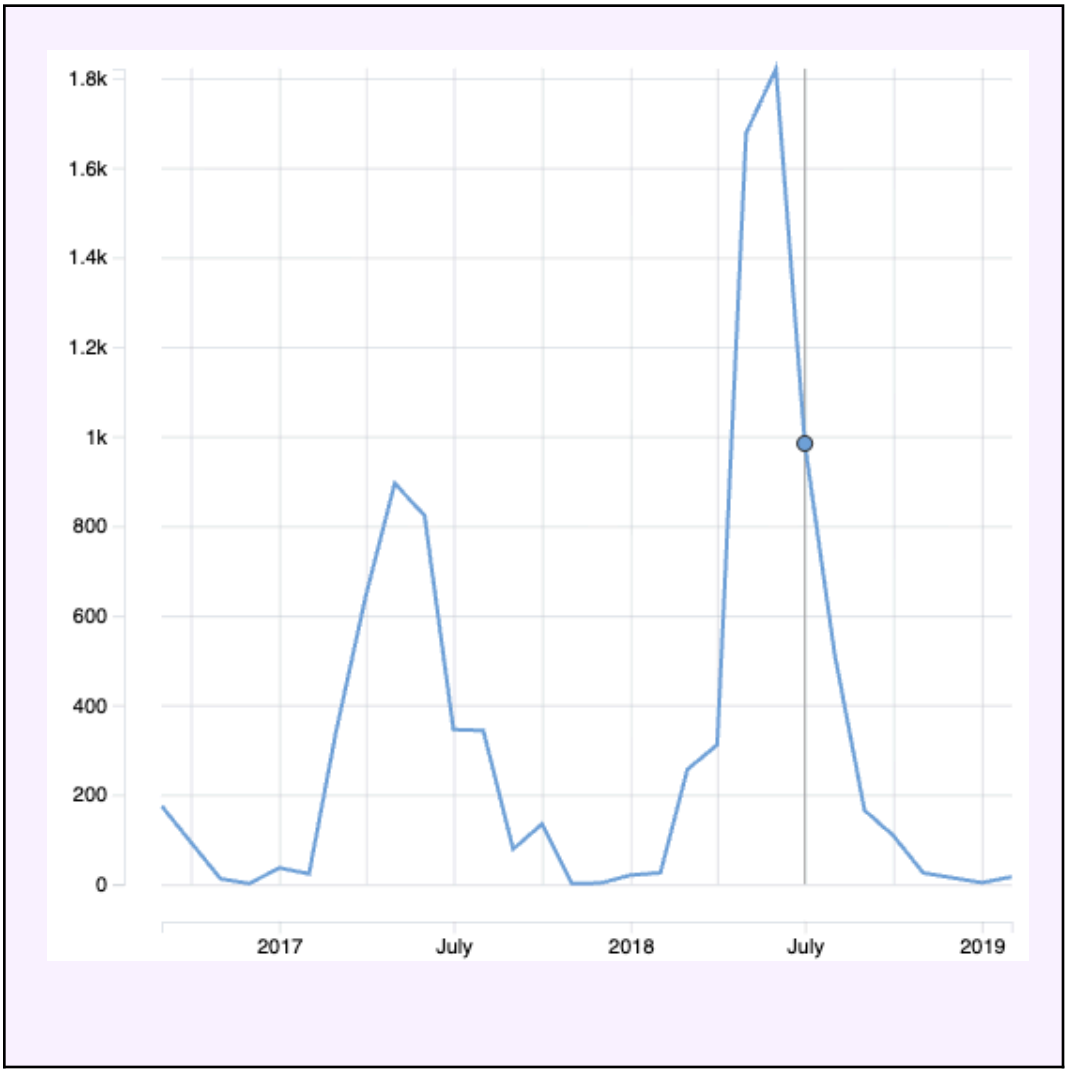
C. The weather data includes values for all 50 states. Modify your query once more so that it only shows information from the states that are seen in the jobs data. Note: Your query should **NOT** use a `JOIN` clause, but instead use a **subquery**!

(paste your query below 👇)

```
SELECT
  date_trunc('month', datetime) AS monthly,
  COUNT(*) as num_weather_events
FROM hover.weather
WHERE date_trunc('month', datetime) >= '2016-09-01'
```

```
  AND state IN (SELECT DISTINCT job_location_region_code FROM
hover.jobs)
GROUP BY
  monthly
ORDER BY
  monthly
```

Visualize the filtered data again with the built-in visualizer and compare the filtered data to the original data from 2B. Did the filtering change anything about the pattern?



The pattern I found of the summer season and winter season correlation is still the same but there are differences in regards to the steepness of how quickly it increases or decreases.

D. Write a few-sentence summary describing the relationship between the job requests and weather events you observed in Tasks 1 and 2.

(write your **answer** below 👇)

Both graphs show that the winter season has an extremely low number of job requests and weather events and the summer season has a high number of job requests and weather events. The weather events graph show a more steep slope suggesting a sudden increase between the winter season and the summer season and a more increase when compared to the job requests graph.

— **Task 3:** Does weather affect demand for jobs?

Given the perceived rel§ ationship between the weather event graph and job requests graph, you will now investigate if there is indeed a relationship between adverse weather events and job requests.

A colleague of yours already wrote a query that returns the total number of weather events grouped at the weekly level for each state in the weather data. You'll build off their work.

A. Write a query that performs a `JOIN` on the `jobs` table and the `weekly_weather_events` table your colleague created. Since you are only interested in the weeks that have had an adverse weather event, use an `INNER`

`JOIN` to match the tables on both the week timestamp AND the state. If you don't do this, you will end up with output crossing states, e.g. requests in TX incorrectly joined to events in KS.

`SELECT` the following columns from the jobs table in your query:

- **`job_deliverable`,**
- **`job_location_region_code`,**
- **`job_first_upload_complete_datetime`**, truncated down to the 'week' level. Alias this column as **`job_ts`**.

And from the `weekly_weather_events` table, you will need the following:

- **`n_weather_events`**

Remember to use the function and NOT the alias **`job_ts`** when joining to the `weather_ts` **`column`**, otherwise you will get an error. If done correctly, your output table will have 25,257 rows.

(paste your query below 👇)

```
SELECT
  j.job_deliverable,
  j.job_location_region_code,
  date_trunc('week', j.job_first_upload_complete_datetime) AS
Week,
  w.n_weather_events
FROM
  hover.weekly_weather_events AS w
INNER JOIN hover.jobs as j
  ON w.state = j.job_location_region_code
     AND w.weather_ts = date_trunc('week',
j.job_first_upload_complete_datetime)
```

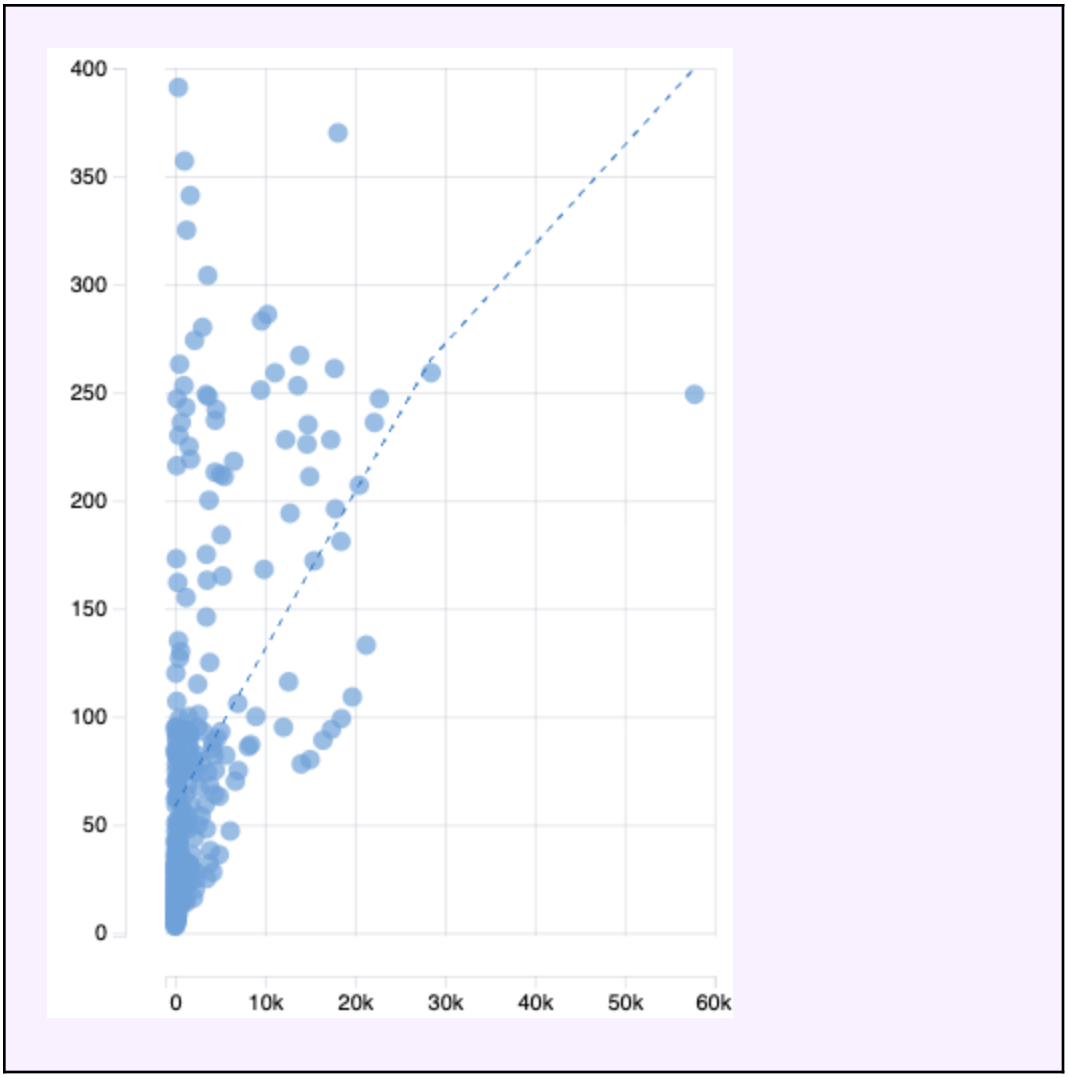
B. Use your query in 3A as a subquery in a new query that counts the total number of jobs and the total number of weather events for each state and

week. The variable `job_ts` can be used to count the total number of jobs. Order your output alphabetically by state. If done correctly, your output table should have 320 rows.

(paste your query below 👇)

```
SELECT
  State,
  Week,
  Count(*) AS num_jobs,
  SUM(n_weather_events) AS total_weather_events
FROM
  (SELECT
    j.job_deliverable,
    j.job_location_region_code AS State,
    date_trunc('week', j.job_first_upload_complete_datetime)
AS Week,
    w.n_weather_events
  FROM
    hover.weekly_weather_events AS w
  INNER JOIN hover.jobs as j
    ON w.state = j.job_location_region_code
       AND w.weather_ts = date_trunc('week',
j.job_first_upload_complete_datetime)
  ) AS temp_table
GROUP BY
  State,
  Week
ORDER BY
  State
```

C. Now you are ready to determine whether there is a relationship between adverse weather events and job requests. Using the built-in visualizer, create a scatterplot of your data. On your x-axis you should have the total number of weather events and on the y-axis you should have the total number of job requests. Check the box for “show trendline”.



Is there a relationship between adverse weather events and job requests?

(write your **answer** below 👇)

The scatterplot shows a positive relationship. There seems to be more wider ranges of data to begin with which shows a weak positive relationship and this suggests that there are other factors other than weather events contributing to the number of job requests but as the weather events increase its a more steady positive relationship with less scatter amongst the data points

which suggest that weather events still play a role in the number of job requests.

D. You'll consider other factors that could contribute to spikes in job requests. Your goal is to identify additional factors influencing construction activity during extreme weather events.

Try this prompt: What other factors might influence the connection between extreme weather and construction job demand, and what additional data might we want to examine to better understand this relationship?

Based on ChatGPT's response, what's one additional factor that might explain spikes in job requests?

(write your **answer** below 👇)

Other factors influencing the number of job requests may be legal factors such as government policies and possibly government aid to stabilise the economy after a nasty weather event. It could be a regular trend in the job requests for construction employment due to the weather in the particular region. Sometimes, if an entire city needs rapid development, it would need a high number of specialised construction workers.

Possible data needed could be options where suggests how much development was needed at a time or how much buildings when to ruin after a weather event

— Submission

Great work completing this Milestone! To submit your completed Milestone, you will need to download / export this document as a PDF and then upload it to the Milestone submission page. You can find the option to download as a PDF from the File menu in the upper-left corner of the Google Doc interface.